Beyond the Agent Object: Entity Component Systems as an Architecture for Agent-Based Modeling

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Abstract

Agent-based models are commonly implemented around agent objects that combine identity, state, and behavior. Although this organization mirrors the intuitive description of autonomous agents, it makes the complete agent representation the principal unit of model construction and can obscure the population-level processes through which agents interact. We examine Entity Component Systems (ECS) as an alternative architecture in which entities supply identity, components represent state and roles, and systems transform populations selected by component queries.

The comparison addresses three questions. First, how does ECS affect the representation and modification of overlapping and changing roles relative to an idiomatic agent-centered implementation of the same model? Second, how does system organization affect the locality, reuse, substitutability, and inspectability of population mechanisms? Third, how do dependencies and phase boundaries define information visibility and update timing, and under what conditions do alternative execution orders preserve outcomes?

The comparative case is the single-resource Sugarscape wealth-distribution model introduced by J. M. Epstein and R. L. Axtell [1]. Heterogeneous citizens move and harvest on a regenerating landscape, metabolize sugar, accumulate wealth, age, and die. Optional reproduction and disease add overlapping and changing roles: sex is represented by exclusive tags, infection by a component added on transmission and removed on recovery, and fertility by predicates over age, wealth, sex, and neighborhood state. Movement is implemented under both shuffled-sequential and staged synchronous semantics, making the consequences of observation and commitment timing explicit.

The evaluation pairs the ECS model with a semantics-matched, idiomatic agent-centered reference and separates architecture treatments from schedule treatments. Storage layout, cache behavior, and multicore scaling are assessed as secondary engineering consequences rather than as a separate research question. The paper thereby treats ECS not as an automatic route to parallelism or superior performance, but as an architecture that can make state composition, population mechanisms, and schedule commitments explicit and testable.

Bibliography

[1] J. M. Epstein and R. L. Axtell, *Growing Artificial Societies: Social Science from the Bottom Up*. Washington, DC and Cambridge, MA: Brookings Institution Press, MIT Press, 1996. Accessed: Aug. 13, 2026. [Online]. Available: https://www.brookings.edu/books/growing-artificial-societies/